

## **Freight Rate Prediction Challenge**

### **Machine Learning Engineer Assessment Report**

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**Role: Machine Learning Engineer**

#### **1. Project Overview**

##### **Objective**

The objective of this project was to develop an end-to-end machine learning solution capable of predicting freight transportation rates (posted\_rate) using historical shipment data.

Accurate freight rate prediction enables logistics companies to estimate transportation costs more efficiently, improve pricing decisions, reduce manual effort, and support better operational planning.

The project covers the complete machine learning workflow, including data exploration, preprocessing, feature engineering, model development, evaluation, model selection, and prediction generation for unseen datasets.

#### **2. Dataset Description**

The following datasets were provided as part of the assessment:

| <b>Dataset</b>                      | <b>Purpose</b>                                                                                          |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| train-test.csv                      | Historical shipment data containing the target variable (posted_rate) used for training and evaluation. |
| validation.csv                      | Unseen shipment data requiring predicted freight rates.                                                 |
| validation-predictions-template.csv | Template used to generate the final validation submission file.                                         |
| december-chart-inputs.csv           | Shipment records used to generate December freight rate predictions.                                    |

The target variable for prediction was:

posted\_rate

### **3. Exploratory Data Analysis (EDA)**

Initial exploration of the datasets was performed to understand their structure and identify preprocessing requirements.

The following observations were made:

The dataset contained both numerical and categorical variables.

Missing values were present in the weight and market\_index columns.

Shipment information included pickup location, delivery location, equipment type, shipment weight, distance, market index, and shipment date.

The posted\_rate column served as the regression target.

Date information provided opportunities to create additional predictive features.

The insights obtained during exploration guided the preprocessing and feature engineering steps.

### **4. Data Preprocessing**

To improve data quality and prepare the datasets for machine learning, several preprocessing techniques were applied.

Missing Value Handling

Missing values in:

Weight

Market Index

were replaced using the median values calculated from the training dataset.

Median imputation was selected because it is less sensitive to extreme values than the mean.

## **Date Processing**

The shipment date was converted into datetime format, and additional features were extracted:

Month

Day

Day of Week

Quarter

Week Number

These features enable the models to capture seasonal and temporal patterns.

## **Categorical Feature Encoding**

Two encoding techniques were applied:

Frequency Encoding

Applied to:

Pickup location

Delivery location

Locations appearing more frequently in the training data received larger frequency values.

## **One-Hot Encoding**

Applied to:

Equipment Type

This transformed categorical equipment values into numerical binary variables suitable for machine learning algorithms.

## **Feature Alignment**

To ensure consistency across datasets, all training, validation, and December datasets were aligned so they contained identical feature columns before model training and prediction.

## **5. Feature Engineering**

Several additional features were created to improve predictive performance.

The final feature set included:

Distance

Weight

Market Index

Month

Day

Day of Week

Quarter

Week Number

Frequency-Encoded Pickup Location

Frequency-Encoded Delivery Location

One-Hot Encoded Equipment Type

These engineered features provided additional information that helped the models better understand freight pricing patterns.

## **6. Training and Validation Strategy**

The labeled dataset was divided into:

85% Training Data

15% Testing Data

using Scikit-learn's `train_test_split()` function with a fixed random state (`random_state=42`) to ensure reproducibility.

The unseen validation dataset did not contain target values and was therefore used only for generating final predictions after selecting the best-performing model.

## 7. Machine Learning Models Evaluated

Five regression algorithms were trained and compared:

Linear Regression

Random Forest Regressor

Gradient Boosting Regressor

XGBoost Regressor

CatBoost Regressor

Each model was trained using the same training dataset and evaluated using identical testing data to ensure a fair comparison.

## 8. Model Evaluation

Model performance was evaluated using:

$R^2$  Score

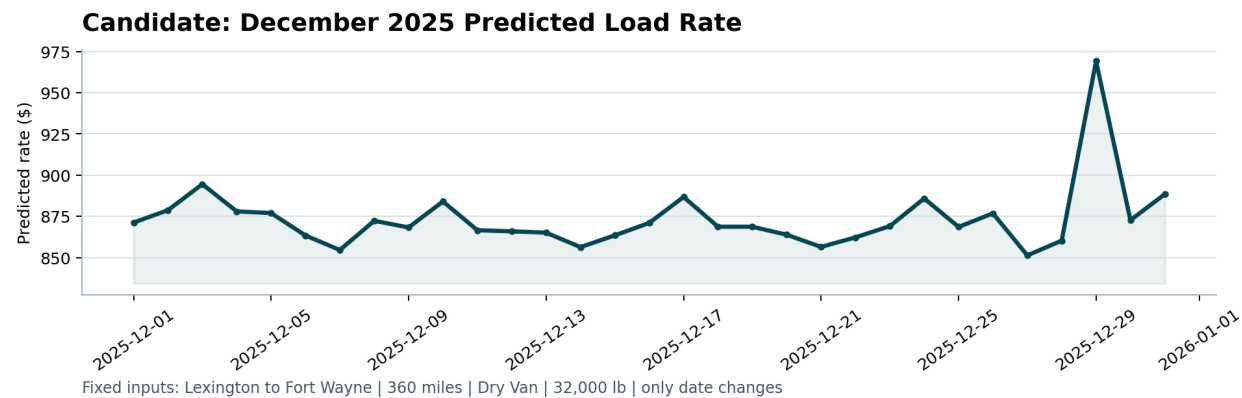
Mean Squared Error (MSE)

The comparison results are shown below.

| Model             | $R^2$ Score | MSE        |
|-------------------|-------------|------------|
| CatBoost          | 0.856666    | 317,482.34 |
| Gradient Boosting | 0.855791    | 319,418.38 |

|                   |          |            |
|-------------------|----------|------------|
| Linear Regression | 0.852259 | 327,243.16 |
| XGBoost           | 0.848560 | 335,435.36 |
| Random Forest     | 0.837174 | 360,655.07 |

CatBoost achieved the highest  $R^2$  score and the lowest Mean Squared Error, making it the best-performing model for this dataset.



candidate\_december.png

## 9. Final Model Selection

CatBoost Regressor was selected as the final model because it consistently outperformed the other regression algorithms during evaluation.

CatBoost is particularly effective for structured tabular datasets because it captures nonlinear relationships between features while reducing overfitting through gradient boosting.

Its superior performance on the testing dataset demonstrated its ability to generalize well to unseen shipment records.

## 10. Prediction Generation

The trained CatBoost model was used to generate predictions for:

validation.csv, producing validation\_predictions.csv

december-chart-inputs.csv, producing december\_predictions.csv

Both prediction files were generated in the required submission format.

## **11. Challenges Encountered**

Several practical challenges were addressed during the project.

### **Missing Data**

Missing values in the weight and market index columns were handled using median imputation.

### **Feature Consistency**

Different datasets contained slight structural differences.

This issue was resolved by aligning feature columns before prediction.

### **Categorical Variables**

Machine learning algorithms require numerical input.

Frequency encoding and one-hot encoding were used to convert categorical variables into numerical representations.

### **Model Selection**

Multiple regression algorithms were trained and compared to identify the most accurate solution rather than relying on a single model.

## **12. Limitations**

The current implementation has several limitations.

Hyperparameter tuning was limited, and default or manually selected parameters were used.

Model evaluation was performed using a single train-test split instead of k-fold cross-validation.

Only the provided shipment attributes were used. External information such as fuel prices, weather conditions, traffic, or economic indicators could potentially improve prediction accuracy.

## **13. Future Improvements**

If additional time were available, the following improvements would be explored:

Hyperparameter optimization using GridSearchCV or Optuna.

K-fold cross-validation to improve evaluation robustness.

Feature importance analysis to better understand the most influential variables.

Additional feature engineering, including seasonal and route-based features.

Ensemble techniques combining multiple high-performing models.

Incorporation of external logistics and market data.

## **14. Conclusion**

This project successfully implemented a complete machine learning pipeline for freight rate prediction.

The workflow included data exploration, preprocessing, feature engineering, model development, evaluation, model comparison, and prediction generation.

Five regression algorithms were evaluated, with CatBoost Regressor achieving the best overall performance ( $R^2 = 0.856666$ ,  $MSE = 317,482.34$ ). Based on these results, CatBoost was selected to generate the final prediction files for both the validation and December datasets.

The final solution demonstrates a practical end-to-end machine learning workflow and provides a reliable approach for predicting freight transportation rates using historical shipment data.

## **Author**

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**Machine Learning Engineer Assessment**